Computer Aided Manufacturing

Laurea Magistrale (Master of Science) in
Automation and Control Engineering

1 year, 1 semester

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Learning Targets

This course introduces the student to the fundamentals of **manufacturing processes** used in **discrete-part fabrication** along with their **needs** in terms of **automation** and **computer-based intelligence**, according to the smart manufacturing paradigm.

Main Topics:

1. Introduction to Manufacturing
2. Industrial Geometrical Metrology
3. Manufacturing Processes
4. CNC Machine Tools and Computer-Aided Manufacturing

LEARNING TARGETS

Lectures and other didactic activities will allow students to:

- know the structure and behaviour of **engineering materials**;
- understand the basic design requirements of a product, and in particular **geometrical product specification and verification**;
- know the basics of manufacturing and **transformation theory**;
- understand the **classification** of manufacturing processes and systems;
- know **the basic principles, tools, and machines** of the considered processes;
- **design feasible manufacturing and inspection processes** for a product;
- know the computer numerical control (CNC) and the **CNC machines**;
- understand the needs and the benefits of the adoption of **CAD/CAM/CAPP** techniques;
- understand the **need for automation and computer-based intelligence** in manufacturing.

Learning Targets

With respect to the study programme, the course will allow to reach the following generic learning outcomes (according to the Dublin Descriptors):

- capability of knowledge and understanding,
- capability of applying knowledge and understanding,
- capability of making judgements,
- capability of communication.

The students should have knowledge related to:

- calculus and geometry;
- the basics of physics and mechanics;
- the basics of measurements;
- the basics of automation and control engineering.

FINAL GRADING

The final grading will be based on **a written test and an oral examination**.

The **written test** will consist of:

- *Four numerical problems to be solved in 2 hours* aimed at highlighting the student's ability to reason and apply knowledge on simple industrial examples (capability of applying knowledge and understanding). The MS Forms will be used to report numerical results and the grade will be based on these numerical results.

The **oral examination** will consist of:

- open questions on the basic topics of the course (capability of knowledge and understanding, capability of communication);
- open questions aimed at highlighting the student's ability to reason and develop links among the different topics of the course (capability of applying knowledge and understanding, of making judgements, and of communication).

Final Grading

Only for the **first exam** of the winter session of the A.Y., the written test will be divided into an intermediate written test (midterm, November) and a final written test (first exam date, January).

Both **intermediate and final written test** will consist of *three numerical problems to be solved in 1,5 hours* aimed at highlighting the student’s ability to reason and apply knowledge on simple industrial examples (capability of applying knowledge and understanding). The MS Forms will be used to report numerical results and the grade will be based on these numerical results.

The final written test can only be taken if the midterm grade is sufficient. If the student fails the midterm, he/she can take the full exam from the earliest available date in January.

If the student passes both the intermediate and final written test, he/she must take the final oral exam during the first exam session (that of the final written test).

LEARNING MATERIAL

Mikell P. Groover
Principles of Modern Manufacturing: Materials, Processes, and Systems
John Wiley and Sons, 2016

Robert J. Hocken & Paulo H. Pereira
Coordinate Measuring Machines and Systems
CRC Press, 2012

Mikell P. Groover
Automation, Production Systems, and Computer-Integrated Manufacturing
Pearson, 2015, Chapter 6 and Chapter 7

David Dornfeld and Dae-Eun Lee
Precision Manufacturing
Springer, 2008, eBook, Chapter IX

WeBeep Website: You are responsible for checking the class website at least on a weekly basis.

CLASS ORGANIZATION

Instructional Mix

- Lectures (~ 70 h)
- Class work (~ 30 h)
- Tutoring (20 h)

Class Organization:

- in presence (face mask recommended)
- no streaming (streaming will be available on September 23 and 26)
- recording will always be available

Class Schedule

Monday	15:15 – 17:15	B8.0.2
Tuesday	13:15 – 15:15	B8.0.2
Thursday	11:15 – 13:15	L.03
Friday	13:15 – 15:15	L.15

Q&A